

17-2a The Prisoners' Dilemma

The prisoners' dilemma is a story about two criminals who have been captured by the police. Let's call them Bonnie and Clyde. The police have enough evidence to convict Bonnie and Clyde of the minor crime of carrying an unregistered gun, so that each would spend a year in jail. The police also suspect that the two criminals have committed a bank robbery together, but they lack hard evidence to convict them of this major crime. The police question Bonnie and Clyde in separate rooms and offer each of them the following deal:

"Right now, we can lock you up for 1 year. If you confess to the bank robbery and implicate your partner, however, we'll give you immunity and you can go free. Your partner will get 20 years in jail. But if you both confess to the crime, we won't need your testimony and we can avoid the cost of a trial, so you will each get an intermediate sentence of 8 years."

If Bonnie and Clyde, heartless bank robbers that they are, care only about their own individual sentences, what would you expect them to do? [Figure 1](#) shows the *payoff matrix* for their choices. Each prisoner has two strategies: confess or remain silent. The sentence each prisoner gets depends on the strategy he or she chooses and the strategy chosen by his or her partner in crime.

Figure 1

The Prisoners' Dilemma

In this game between two criminals suspected of committing a crime, the sentence that each receives depends both on his or her decision whether to confess or remain silent and on the decision made by the other.

 A two by two grid decision box (four quadrants) lists the sentences that will result from the decisions Bonnie and Clyde could individually take regarding making a confession. Bonnie's decision is on the top (Confess is on the left and Remain Silent on the right) and Clyde's decision is on the left (Confess on the top and Remain Silent on the bottom). If both confess (upper left quadrant), each gets 8 years. If Bonnie remains silent while Clyde confesses (upper right quadrant), Bonnie gets 20 years while Clyde goes free. If Clyde remains silent while Bonnie confesses (lower left quadrant), Clyde gets 20 years while Bonnie goes free. If both remain silent (lower right quadrant), each gets 1 year.

Consider first Bonnie's decision. She reasons as follows: "I don't know what Clyde is going to do. If he remains silent, my best strategy is to confess, because then I'll go free rather than spending a year in jail. If he confesses, my best strategy is still to confess, because then I'll spend 8 years in jail rather than 20. So, regardless of what Clyde does, I am better off confessing."

In the language of game theory, a strategy is called a **dominant strategy** (a strategy that is best for a player in a game regardless of the strategies chosen by the other players) if it is the best strategy for a player to follow regardless of the strategies pursued by other players. In this case, confessing is a dominant strategy for Bonnie. She spends less time in jail if she confesses, regardless of whether Clyde confesses or remains silent.

Now consider Clyde's decision. He faces the same choices as Bonnie, and he reasons the same way. Regardless of what Bonnie does, Clyde can reduce his jail time by confessing. In other words, confessing is also a dominant strategy for Clyde.

In the end, both Bonnie and Clyde confess, and both spend 8 years in jail. This outcome is a Nash equilibrium: Each criminal is choosing the best strategy available given the strategy the other is following. Yet, from their standpoint, the outcome is terrible. If they had *both* remained silent, both of them would have been better off, spending only 1 year in jail on the gun charge. Because each pursues his or her own interest, the two prisoners together reach an outcome that is worse for each of them.

You might have thought that Bonnie and Clyde would have foreseen this situation and planned ahead. But even with advanced planning, they would still run into problems. Imagine that, before the police captured Bonnie and Clyde, the two criminals had agreed not to confess. Clearly, this pact would make them both better off *if* they both lived up to it because each would spend only 1 year in jail. But would the two criminals in fact remain silent simply because they had agreed they would? Once they are being questioned separately, the logic of self-interest takes over and leads them to confess. Cooperation between the two prisoners is difficult to maintain, because cooperation is individually irrational.